

Bank Marketing Campaign Analysis Report

Executive Summary

The objective of this project was to analyze customer data from a bank marketing campaign and predict whether a customer would subscribe to a term deposit. Using data analytics and machine learning techniques, key customer behaviors and campaign factors were identified to improve marketing effectiveness.

Key Findings

- The dataset contained over 41,000 customer records with 21 features.
- Only 11.27% of customers subscribed to the term deposit, indicating a highly imbalanced dataset.
- Call duration was the most influential factor affecting customer subscription decisions.
- Economic indicators such as Euribor interest rates and employment levels significantly impacted customer responses.
- Previous successful campaign interactions increased the probability of future subscriptions.

Model Performance

Multiple machine learning models were developed and evaluated, including Logistic Regression, Random Forest, and XGBoost. The Random Forest model delivered the best performance with approximately 91% prediction accuracy.

Business Impact

The analysis enables the bank to:

- Identify high-potential customers.
- Improve campaign targeting strategies.
- Increase marketing conversion rates.
- Reduce operational costs through data-driven decision making.

Recommendations

1. Focus on customers with a history of positive campaign responses.
2. Improve call quality and customer engagement.
3. Use predictive analytics to prioritize leads.
4. Optimize campaign frequency to avoid customer fatigue.

Conclusion

The project successfully demonstrated how data analytics and machine learning can support banking institutions in improving marketing campaign effectiveness and customer acquisition strategies.